

ROLL NO:250701573

EXPERIMENT NO:15 DATA COLLECTION AND INITIAL EXPLORATION

PROGRAM:

```
import pandas as pd
```

```
db = pd.read_csv("diabetes_modified.csv")
```

```
print(db.head())
```

```
print(db.info())
```

```
print(db.describe())
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
db.hist(bins=50, figsize=(20,15))
```

```
plt.show()
```

```
sns.pairplot(db)
```

```
plt.show()
```

OUTPUT:

Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	\
0	2	118	72	30	90	32.1
1	5	132	80	35	125	35.4
2	1	95	66	23	80	27.8
3	8	168	88	40	210	38.5
4	3	110	70	28	100	30.2

	DiabetesPedigreeFunction	Age	Outcome
0	0.45	29	0
1	0.62	41	1

2	0.31	23	0
3	0.78	52	1
4	0.40	31	0

<class 'pandas.DataFrame'>

RangeIndex: 20 entries, 0 to 19

Data columns (total 9 columns):

#	Column	Non-Null Count	Dtype
0	Pregnancies	20 non-null	int64
1	Glucose	20 non-null	int64
2	BloodPressure	20 non-null	int64
3	SkinThickness	20 non-null	int64
4	Insulin	20 non-null	int64
5	BMI	20 non-null	float64
6	DiabetesPedigreeFunction	20 non-null	float64
7	Age	20 non-null	int64
8	Outcome	20 non-null	int64

dtypes: float64(2), int64(7)

memory usage: 1.5 KB

None

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin \
count	20.000000	20.000000	20.000000	20.000000	20.000000
mean	4.400000	131.400000	75.850000	31.350000	138.650000
std	2.81724	24.963974	8.041177	6.326843	54.816704
min	0.000000	95.000000	62.000000	20.000000	70.000000
25%	2.000000	111.500000	69.750000	26.750000	89.500000
50%	4.000000	127.500000	76.500000	31.500000	130.000000
75%	6.250000	151.250000	81.250000	36.250000	186.250000
max	10.000000	178.000000	90.000000	42.000000	240.000000

	BMI	DiabetesPedigreeFunction	Age	Outcome
count	20.000000	20.000000	20.000000	20.000000
mean	32.615000	0.525500	38.600000	0.500000
std	4.103692	0.207909	12.076075	0.512989
min	25.900000	0.220000	22.000000	0.000000
25%	29.575000	0.352500	28.750000	0.000000
50%	32.450000	0.500000	37.000000	0.500000
75%	35.600000	0.715000	47.500000	1.000000
max	40.100000	0.890000	61.000000	1.000000



